

Kelvin Chen

CONTACT INFORMATION	Medical Research Bldg 4 409 Lane Rd Charlottesville, VA 22903	Phone: (xxx) xxx-xxxx Email: ddw4hp@virginia.edu Homepage: https://k9chen.github.io
RESEARCH INTERESTS	Translational and clinical neuroscience, neuroimmunology, cerebrovascular physiology, neuropathology, neuropharmacology, neurosurgery	
EDUCATION	University of Virginia , Charlottesville, VA BA, Neuroscience; Minor, Bioethics • Advisor(s): Jianjie Ma, PhD ; Ling Qi, PhD • Thesis: <i>MG53/CitH3 modulation attenuates NETosis-mediated neuroinflammation</i> Virginia Peninsula Community College , Hampton, VA Dual Enrollment, Biological Science & Mathematics	Aug 2022 – May 2026 Sept 2020 – Jun 2022
RESEARCH EXPERIENCE	UVA School of Medicine , Charlottesville, VA <i>Undergraduate Researcher, Dept. of Surgery</i> Advisor: Jianjie Ma, PhD • Treatment of cerebral ischemia with CHO cell-derived rhMG53 <i>Undergraduate Researcher, Dept. of Neurosurgery</i> Advisor: Petr Tvrdik, PhD • Recombinase-based spatiotemporal mapping of myeloid cells <i>Undergraduate Researcher, Dept. of Neuroscience</i> Advisor: Lulu Jiang, MD, PhD • Proximity labeling of translocated RNA with TurboID in iPSCs <i>Undergraduate Researcher, Dept. of Pharmacology</i> Advisor: Julius Zhu, PhD • Image analysis with genetically encoded fluorescent sensors Barrow Neurological Institute , Phoenix, AZ <i>Research Intern, Dept. of Translational Neuroscience</i> Advisor: Richard Dortch, PhD • Numerical simulation of multi-compartmental dMRI models <i>Research Intern, Dept. of Translational Neuroscience</i> Advisor: Richard Dortch, PhD • Automatic segmentation of peripheral nerve histology images Hampton University , Hampton, VA <i>Research Intern, Dept. of Chemistry & Biochemistry</i> Advisor: Peter Njoki, PhD • Viral kinetic ODE-based modeling of antigen-conjugated NPs	Dec 2024 – Present Jan 2024 – Dec 2024 Aug 2023 – Jan 2024 Sep 2022 – May 2023 May 2024 – Aug 2024 May 2023 – Aug 2023 Sept 2021 – May 2022
AWARDS & GRANTS	Ingrassia Family Research Award (\$2,000), UVA Harrison Undergraduate Research Award (\$5,000), UVA Echols Scholarship (top 5% of class), UVA Dean’s List of Distinguished Students, UVA University Achievement Award Scholarship (full-tuition), UVA	2025 2025 2023 2023 – 2025 2022 – 2026

PUBLICATIONS &
PRESENTATIONS

 GOOGLE SCHOLAR

- [C.1] Ketsiri, T., **Chen, K.**, Xu, J., Dortch, R. D. *Numerical validation of multi-compartment diffusion biomarkers of peripheral nerve trauma*. 33rd Annual Meeting & Exhibition of the International Society for Magnetic Resonance in Medicine, Honolulu, HI, May 2025. [[Abstract](#)]
- [C.2] Ketsiri, T., **Chen, K.**, Xu, J., Dortch, R. D. *Validation of multi-compartmental diffusion MRI models for peripheral nerve trauma*. Gordon Research Conference on In Vivo Magnetic Resonance, Andover, NH, Jul 2024. [[Abstract](#)]
- [O.1] **Chen, K.**, Lee, K. E., Park, K. H., Ma, J. *MG53/CitH3 modulation attenuates NETosis-mediated neuroinflammation*. UVA Fozdar Symposium for Undergraduate Neuroscience Research, Charlottesville, VA, Apr 2026. [[Slides](#)]
- [P.1] **Chen, K.**, Lee, K. E., Park, K. H., Ma, J. *Co-targeting MG53 and CitH3 attenuates cerebral ischemia-reperfusion injury*. UVA Undergraduate Research Symposium, Charlottesville, VA, Apr 2026. [[Poster](#)]
- [P.2] **Chen, K.**, Lee, K. E., Park, K. H., Ma, J. *Co-targeting MG53 and CitH3 attenuates cerebral ischemia-reperfusion injury*. UVA Fozdar Poster Session, Charlottesville, VA, Dec 2025. [[Poster](#)]
- [P.3] **Chen, K.**, Ketsiri, T., Dortch, R. D. *Spherical mean diffusion-weighted MRI reveals peripheral axonal pathology following trauma*. BNI Undergraduate Research Symposium, Phoenix, AZ, Aug 2024. [[Poster](#)]
- [P.4] **Chen, K.**, Sadrabadi, M. S., Dortch, R. D. *Geometry-informed multi-compartmental diffusion MRI modeling of sciatic nerve crush*. BNI Undergraduate Research Symposium, Phoenix, AZ, Aug 2023. [[Poster](#)]

TEACHING
EXPERIENCE

University of Virginia, Charlottesville, VA

Teaching Assistant, Dept. of Chemistry

CHEM 1810 Principles of Chemical Structure (Accelerated) F24, F25

CHEM 2321 Organic Chemistry II Laboratory S24, S25

CHEM 2311 Organic Chemistry I Laboratory F23, F24, F25

CHEM 1811 Principles of Chemical Structure Laboratory (Accelerated) F23

Teaching Assistant, Dept. of Physics

PHYS 2030 Principles of Physics I Laboratory (Pre-Health) F25

PHYS 1429 Introductory Physics I Laboratory (Engineering) S25

Teaching Assistant, Dept. of Psychology

PSYC 3210 Research Methods: Psychobiology Laboratory S24

SERVICE &
OUTREACH

UVA Office of Citizen Scholar Development, Charlottesville, VA

Symposium Volunteer

Apr 2023, Apr 2024

- UVA Undergraduate Research Symposium

UVA School of Medicine, Charlottesville, VA

Symposium Volunteer

Mar 2025

- Advancing Innovations in Translational Research into Human Applications Symposium in Honor of Hiroshi Takeshima

W. M. Keck Center for Cellular Imaging, Charlottesville, VA

Workshop Volunteer

Mar 2024

- 21st Annual Workshop on FLIM and FRET Microscopy and Metabolic Imaging

OTHER	Outreach Ministry Volunteer, Grace Christian Fellowship	2023 – Present
	Senior Associate, Blossom Together	2023 – Present
	Editor, Grounds: The Virginia Journal of Bioethics	2023 – Present
	Senior Translator, The Cavalier Daily	2023 – Present
	Editor, Oculus: The Virginia Journal of Undergraduate Research	2022 – Present
	Investigator, University Judiciary Committee	2022 – Present
	Health & Medical Services Volunteer, Madison House	2022 – 2023
AFFILIATIONS	American Neurological Association	Jun 2024 – Present
SKILLS	Programming: Python, MATLAB, R, C/C++, HTML/CSS, JavaScript, L ^A T _E X Frameworks: PyTorch, TensorFlow, NumPy, Pandas, Scikit-learn, OpenCV Languages: English, Mandarin	
REFERENCES	Jianjie Ma, PhD Professor & Director Division of Surgical Sciences Dept. of Surgery UVA School of Medicine qga8zx@virginia.edu	Richard Dortch, PhD Associate Professor, Director, & Chair Neuroimaging Innovation Center Dept. of Translational Neuroscience Barrow Neurological Institute richard.dortch@barrowneuro.org
	Ammasi Periasamy, PhD Professor, Director, & Founder W. M. Keck Center for Cellular Imaging Dept. of Biology & Biomed. Eng. University of Virginia ap3t@virginia.edu	Andreas Gahlmann, PhD Associate Professor Center for Membrane & Cell Physiology Dept. of Mol. Physiol. & Biol. Phys. UVA School of Medicine ag5vu@virginia.edu